Assignment 3

This is an individual assignment. This assignment is marked out of 5 points.

Due Date: October 30 Friday, 2026, 11:59PM

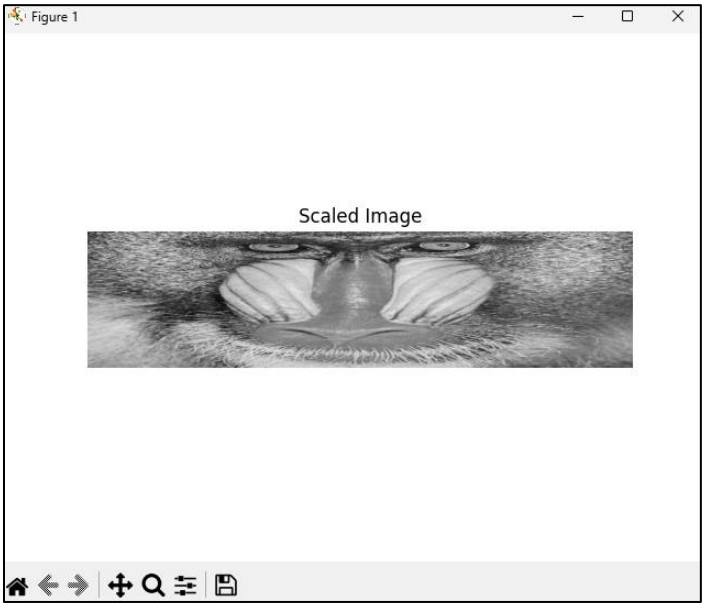
1. Affine transformation:
   If interpolation is needed, use nearest-neighbor interpolation.

   (a) [2 points] Write a function $imageScaling4e(f, cx, cy)$, where f is a grayscale image and cx and cy are positive scaling factors along the x (horizontal) direction and y (vertical) directions. The function must return the scaled image. Test your function by scaling image "BaboonGray.png" to one-half its height and twice its width. Display the result. **You are NOT allowed to use any existing affine-transformation functions, including but not limited to OpenCV helpers such as cv2.warpAffine, cv2.getAffineTransform, cv2.getRotationMatrix2D, or cv2.remap. Implement translations/affine operations manually instead of calling OpenCV affine utilities.**

$$\begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} c_x x \\ c_y y \\ 1 \end{bmatrix}$$

   Hint: The scaling equation is

   Expected output:



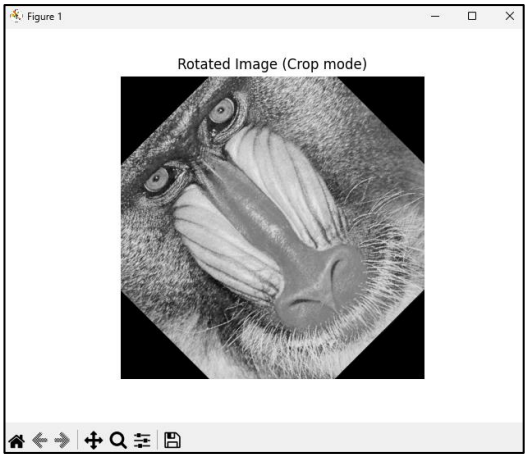
   (b) [3 points] Write a function $imageRotate4e(f, theta, mode)$ for performing image rotation (about the image center), where the $theta$ is the rotation angle in degrees (the

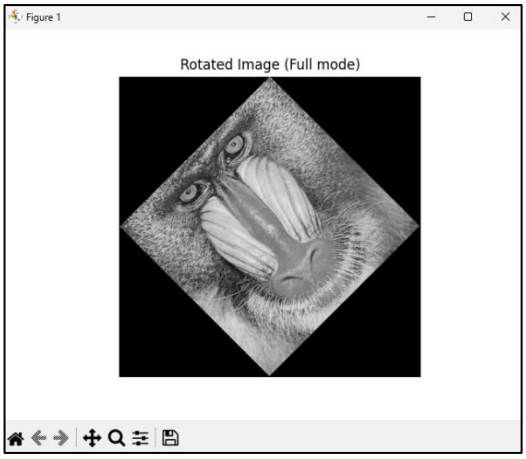
default is zero degrees). A positive angle should produce counterclockwise rotation. If $mode = 'crop'$, the rotated image should be cropped (about its center) to the same size as the input image. This should be the default. If $mode = 'full'$, the output image should be the smallest size capable of containing the full rotated input image for any angle. The background of the rotated image should be black. The function must return the rotated image. To test your function, rotate the image "BaboonGray.png" by 45° using both the $'full'$ and the $'crop'$ modes. Display the results. **You are NOT allowed to use any existing affine-transformation functions, including but not limited to OpenCV helpers such as cv2.warpAffine, cv2.getAffineTransform, cv2.getRotationMatrix2D, or cv2.remap. Implement translations/affine operations manually instead of calling OpenCV affine utilities.**

Hint: The rotation equation is

$$\begin{bmatrix} cos\theta & -sin\theta & 0 \\ sin\theta & cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} xcos\theta - ysin\theta \\ xsin\theta + ycos\theta \\ 1 \end{bmatrix}$$

Expected output:





Requirements:

- Only the following packages are permitted: OpenCV-Python, Matplotlib, NumPy
- OpenCV-Python can only be used for reading the image
- Use Python version 3.10+

Submission:

Submit everything as one zip file to Canvas.